A condition on the multiset of every subblock: OEIS A183589

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Abstract

OEIS A183589 carries an empirical recurrence of order 31 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. The entry's condition on a subblock depends only on which values the block holds and how many times, not on where they sit, so it is decided by the few consecutive lines the block spans. The admissible windows are the vertices of a finite digraph and the arrays counted are exactly its walks; hence $a(n)$ is a walk count on $S = 31$ vertices and is C-finite, and whether the conjectured recurrence annihilates it is settled by a finite exact computation, with the Cayley–Hamilton theorem bounding the work. The entry does not count the arrays themselves but 1/24 of them, so every count below is divided by 24.

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1 The conjecture

OEIS A183589 is "1/24 the number of (n+1) X 5 0..3 arrays with every 2 X 2 subblock containing exactly three distinct values.". It has offset 1 and begins

$$4832, 603766, 74491726, 9207834547, 1137874208678, 140620157014391, 17377952662176322, 2147583158143435$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n)=179*a(n-1)-6470*a(n-2)-111366*a(n-3)+8421556*a(n-4)-27797881*a(n-5)-3641734984*a(n-6)+33282463706*a(n-7)+695892591667*a(n-8)-8771266871058*a(n-9)-59332612133336*a(n-10)+1069043025480468*a(n-11)+1401139251128208*a(n-12)-67977787226742392*a(n-13)+103924226454772488*a(n-14)+2296282350133287328*a(n-15)-7835601773053827936*a(n-16)-39951799309240069504*a(n-17)+217732921395304638464*a(n-18)+281480768985183547392*a(n-19)-3088710469400605992960*a(n-20)+1060118307218454011904*a(n-21)+23235106821777545527296*a(n-22)-29813502870904911822848*a(n-23)-86205910856152217812992*a
> 24)+17634045539691340706611 2*a(n-25)+120352469518364633464832*a(n-26)-4130358369613784494899
> 27)+234127568988368053207 04*a(n-28)+336580017414471294124032*a(n-29)-87482121395567755001856
> 30)-526105267097491734528 00*a(n-31).

As of the "Last modified" line on the live entry (Jun 18 2026 15:34:08, revision 12) this is still recorded as empirical, and nothing on the entry records it as proved.

2 The count is a walk count

The entry counts arrays with 5 columns over the alphabet $\{0, 1, 2, 3\}$, growing one row at a time, and asks that every 2×2 subblock holds exactly 3 distinct values.

That condition does not depend on where in the block a value sits, only on which values the block holds and how many times — it is a function of the block's multiset of entries. In particular it is decided by 2 consecutive rows, the ones the block spans, so carrying the previous 1 row as the state is enough: each new row completes every block that ends on it, and no judgement is left over at the end of the array.

The entry counts not the arrays themselves but 1/24 of them, so every count below is divided by 24; the division comes out exact at every term the entry publishes, which is itself a check on the model.

Merging vertices with identical futures leaves $S = 31$. An admissible array is exactly a walk, so with M the adjacency matrix and ι, τ the vectors recording which states may begin and end an array,

$$a(n) \;=\; \tfrac{1}{24}\,\iota^\top M^{\,n}\tau,$$

the exponent fixed by the entry's own shape and checked against its published terms. In particular a is C-finite.

3 The proof

Lemma 1. *Let $q(t) = t^{31} - \sum_i c_i t^{31-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^{\,j} q(M)\tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^\top M^j q(M)\tau = 0$ for all large j, and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+31} - \sum_i c_i M^{j+31-i}$. For the bound, M^S is an integer combination of $I, M, \ldots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M)\tau$ obey the monic recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. A constant factor in front of a divides out of the residual and changes nothing here. □

Theorem 2.

$$a(n) \;=\; 179\,a(n{-}1){-}6470\,a(n{-}2){-}111366\,a(n{-}3){+}8421556\,a(n{-}4){-}27797881\,a(n{-}5){-}3641734984\,a(n{-}6){+}3$$

for every $n > 30$.

Proof. The vector $w = q(M)\tau$ was formed by 31 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \ldots,$ again exactly; those integers vanish from the index corresponding to $n = 30 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. □

4 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 11 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: the digraph is built by reading the entry's English, and a misreading gives different counts.

Second, the reading was pinned before the digraph was written, by a program that enumerates every array of the given shape one by one, forms each subblock directly and tests the entry's condition on it. That program shares no code with the transfer matrix, so an error in one does not hide an error in the other.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A183589.

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[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.